Toward Creative Agents that Regard Everything as Environment and Tool

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Abstract

In AI, the term agent is often narrowed to systems that invoke APIs or orchestrate a fixed set of predefined skills. This paper instead frames agents as systems that interact with their surroundings while intentionally pursuing goals. Under this view, an **environment** is *any influenceable medium* that accepts actions and returns feedback, and a **tool** is *any resource* whose attributes or affordances can be exploited to advance a goal. We hypothesize that **creative intelligence**, a hallmark of human cognition that remains underdeveloped in AI, emerges when agents move beyond reliance on explicitly specified environments and designer-defined tools. Specifically, creativity arises when agents opportunistically reinterpret their surroundings as interactive media, discovering new affordances and converting ordinary elements of the world into unexpected means for accomplishing their objectives. To support this capability, creative agents should: (i) deliberately select and switch among available environments to probe constraints and uncover opportunities, and (ii) discover and validate tool utility at the level of underlying attributes, rather than through designer-described interfaces.

"Intelligence is what you use when you don't know what to do." —*Jean Piaget*

1. What Is an Agent? The Misuse of a Concept

1.1. Motivation: Overuse and Misinterpretation

The term *agent* is now widely applied to models that can execute API calls or operate software tools. While such systems do take actions, reducing agency to tool invocation is a narrow, technology-dependent view. It treats agents as mere procedure executors within predefined interfaces, rather than entities that reason, adapt, and engage with their surroundings. For example, a customer-support agent restricted to scripted macros may follow the interface perfectly but fail when resolution requires asking clarifying questions, negotiating exceptions, or switching channels (e.g., escalation or phone support). This narrow usage also drifts away from well-established notions of agency in cognitive science, robotics, and decision theory (Barandiaran et al., 2009; Jackson & Williams, 2021; Russell et al., 1995).

True agency is not defined by a fixed set of tools or environments. In everyday problem solving, humans adapt to the environment rather than rely on prescribed actions: lacking a knife, we may borrow one from a neighbor or repurpose a key to cut open a package. Such behaviors leverage context, social interaction, and physical improvisation. They are not determined by a predetermined "action space," but emerge from being embedded in the world (Baber et al., 2014).

If AI agents are expected to assist in similarly open-ended settings, they must exhibit the same foundations. Merely mapping inputs to outputs here is insufficient: an agent must **interact** with its surroundings to acquire new information, and it must **pursue explicit goals** that guide its behavior. Without both traits, a system remains a predictive model, not an agent (Chen et al., 2025).

1.2. Definition

We therefore propose the following definition:

Definition of an Agent

An agent is a model (e.g., LLM or VLM) endowed with the capacity to interact with its environment and intentionally pursue goals.

Two concepts naturally accompany this definition:

- **Environment:** the external medium that can register and respond to an agent's actions.
- **Tools:** the means through which an agent carries out actions to pursue a goal.

These notions are not meant to denote fixed sets of interfaces or capabilities. Rather than specifying predefined APIs, modalities, or action libraries, they emphasize that

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what matters in agency is not the particular mechanisms a model can access, but the roles these mechanisms play in supporting interaction and goal pursuit. In this sense, environments and tools are conceptual components that help describe how agents act and what enables them to accomplish intended outcomes.

Case: Web Agents. Web agents exemplify these two traits clearly. Their *interaction* occurs with web pages and online platforms (*environment*) through actions such as searching, clicking, and form-filling (*tools*), all in service of user goals like shopping (Yao et al., 2022), travel planning (Xie et al., 2024a), information retrieval (Wei et al., 2025), information aggregation (Reddy et al., 2025b), knowledge acquisition (Reddy et al., 2026), and situation analysis (Reddy et al., 2025a). Early systems such as WebGPT and WebAgent (Nakano et al., 2021; Gur et al., 2024) interacted primarily via textual commands, while newer systems like MindACT, WebGUM, and SeeAct (Deng et al., 2023; Furuta et al., 2024; Zheng et al., 2024) engage multimodal webpages that require perception and grounded action. Even as environments expand and tools shift from code execution to visual grounding, the agentic essence remains the same: purposeful interaction toward task goals.

Case: Game Agents. Game agents operate in virtual worlds whose rules and dynamics form the *environment*, while controls, inventories, skills, or crafting systems serve as *tools* for achieving game objectives such as winning or exploration. Benchmarks like GameBench and SmartPlay (Costarelli et al., 2024; Wu et al., 2023) assess an agent's ability to *interact* by planning and adapting through feedback, and to *pursue goals* that require long-horizon reasoning. Broader testbeds such as Orak and lmgame-Bench (Park et al., 2025; Hu et al., 2025) enable this interaction across many games, while others stress generalization through procedural or adversarial environments (Paglieri et al., 2024; Li et al., 2025; Schipper et al., 2025). Open-ended systems like Voyager (Wang et al., 2024a) further reveal that game tools can evolve as agents expand their own skills, but the agent identity remains grounded in acting within the environment to pursue explicit goals.

Case: Embodied Agents. Embodied agents make the interaction–goal linkage explicit in physical or simulated 3D spaces (*environment*), using sensors, motors, manipulators, or memory systems as *tools* to complete tasks. Benchmarks such as EmbodiedBench and EmbodiedCity (Yang et al., 2025; Gao et al., 2024b) evaluate how agents *interact* through perception, navigation, and manipulation, while BEHAVIOR in Habitat 2.0 (Liu et al., 2022) formalizes transferable *goal structures*. Further evaluations focus on aspects that directly condition interaction for goal pursuit, such as memory (Yadav et al., 2025), safety in physical action (Lu et al., 2025), or retrieval-based policy augmentation (Zhu et al., 2024). Systems like Mobile ALOHA and EmbodiedGPT (Fu et al., 2024; Mu et al., 2023) couple multimodal sensing with action generation so that affordances can be used as tools to fulfill tasks. In these settings, tools and environments vary widely, yet agency remains defined by the same criteria: purposeful goal pursuit enabled by interaction with the world.

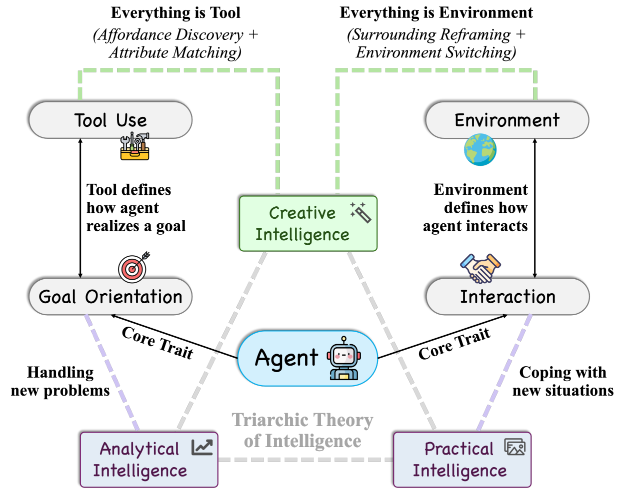

Figure 1. The relations of core concepts: the definition of agents, its relation to triarchic theory of intelligence, and how creativity emerges upon tools and environments.

1.3. Two Perspectives Implicit in the Definition

The definition of agents in the previous section can actually be viewed in reverse, revealing two underlying perspectives that implicitly shape our understanding of agents:

- **Interaction Perspective:** environments define how agents can interact.
- **Goal-Oriented Perspective:** tools define how agents realize a purpose.

These two perspectives are not assumptions added to the concept of an agent, but the consequences of the definition itself. From here, they will guide our reformulation of how creativity should be understood for agents.

2. Why Creativity Matters for Agents

2.1. Missing Dimension of Intelligence

Intelligence in humans emerges when tools are flexibly used (Qin et al., 2024). According to the Triarchic Theory of Intelligence proposed by Robert J. Sternberg, intelligence decomposes into three inter-related aspects: analytical intelligence, practical intelligence, and creative intelligence (Sternberg, 1986). To approach human-level intelligence, an AI agent must demonstrate comparable proficiency across all three dimensions.

When grounded in agents, the three forms of intelligence gain concrete operational meaning. **Analytical intelligence** reflects an agent's academic-style reasoning: the ability to solve structured problems in mathematics, science, programming, or tool-assisted computation. This capability is commonly evaluated through benchmarks targeting the underlying model's reasoning capacity, such as HLE (Phan et al., 2025), OlympiadBench (He et al., 2024), GPQA (Rein et al., 2024), and BFCL (Patil et al., 2025).

Practical intelligence, which in humans represents real-world "street-smarts," corresponds in agents to competence in interactive environments. Rather than operating in isolation, the agent must navigate a dynamic context, make sequential decisions, and adapt its strategies to succeed. This is tested in settings such as game-based evaluation (Qian et al., 2025a), interactive web tasks (Yao et al., 2022), or full operating-system interfaces (Xie et al., 2024b). Together, these benchmarks assess not only whether an agent can solve a task, but whether it can adapt effectively within an environment to achieve its goals.

From the examples above, we observe that many current benchmarks for agent intelligence implicitly test only goal-oriented performance (analytical) and environment interaction (practical), but they largely overlook creative intelligence, whose meaning varies by context and which we analyze in the next section.

2.2. What Counts as Creativity for Different Systems?

Creative intelligence carries distinct meanings depending on the system level. In *humans*, the Triarchic Theory defines it as the ability to use existing knowledge to craft new ways of handling novel problems or coping with new situations (Sternberg, 1986). Transferred to a *model*, "existing knowledge" becomes the parametric knowledge encoded, and creativity means repurposing it for new tasks.

Grounded further in *agent* settings, we argue that creative intelligence links directly to the other two intelligences: "handling new problems" aligns with goal-orientation (analytical intelligence), and "coping with new situations" aligns with rich environment interaction (practical intelligence). Thus creative intelligence naturally builds upon analytical and practical intelligence, but adds a novelty dimension:

> **Definition of Agent Creative Intelligence**
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> Agent creative intelligence builds upon analytical and practical intelligence but emphasizes new ways to achieve goals or new ways to perform interactions with the environment. Concretely, it is the ability to creatively choose environments to interact with and creatively use or invent tools to pursue goals.

Given the central role of goal-directed interaction in our definition of an agent, the absence of creative intelligence is a critical shortcoming. If agents cannot invent novel strategies or reshape their operational context, they will fall short of human-scale intelligence or autonomy. We therefore naturally ask:

How does creativity emerge from interaction, and how does it emerge from goal pursuit?

This question sets the stage for our later inquiry, and at the same time underscores the urgent need to embed creativity into agent architectures if they are to operate robustly in open-ended, dynamic settings.

3. Interaction Perspective: Everything Is Environment

3.1. Generalizing the Concept

Conventional agent research treats an "environment" as a predefined, external space where tasks unfold. Typical examples include web interfaces, operating systems, file systems, or simulated 3D worlds (Yao et al., 2022; Xie et al., 2024b; Gao et al., 2024b). These are *explicit* environments: discretized domains with clearly specified states, allowed actions, and feedback channels.

We argue that this view is too narrow. Instead, an environment should be understood more broadly as:

Anything that can receive an agent's influence by taking in an action and producing a response.

This is supported by that, in human cognition, "environment" often refers to any perceivable and influenceable surrounding (Wilson, 2002; Shapiro, 2019). Under this broader definition, explicit task environments are only one type of possible interactive context. By contrast, existing agents and benchmarks typically assume a single, fixed environment, and constrain evaluation to that interface. This design choice is largely pragmatic: fixed environments make evaluation reproducible and integration manageable, but it also hard-codes where agency is allowed to operate, thus limiting creativity to behaviors within a predefined sandbox. Therefore, we propose:

> **Interaction-Based View of Creative Agency**
>
> A creative agent should treat *all influenceable surroundings* as potential environments, and dynamically decide which of them to interact with.

Recognizing "everything as environment" is fundamental to creativity. Under this view, creativity becomes not only about how an agent acts within an environment, but about how it can choose, repurpose or expand its operational context. In subsequent sections, we will show how such non-traditional environments can serve as ground for agent creativity when treated as interactive spaces.

3.2. Studies of Special Environments

Our generalized view above implies that many practically important contexts are currently under-used as *interactive* environments. Most agent benchmarks still assume a single, fixed sandbox (Yao et al., 2022; Xie et al., 2024b; Fan et al., 2022), and treat everything else including users, other models or other agents as background or infrastructure. In contrast, work in mixed-initiative interaction and human–AI collaboration shows that who or what an agent acts on (user, collaborator, judge, peer) crucially shapes the space of possible behaviors (Horvitz, 1999; Allen et al., 1999). In this section, we highlight three classes of "special environments" that become central once interaction is taken seriously through case studies.

Case: The User as Environment. Under our broad definition, a user is an environment, where the agent can take actions through questions, suggestions, or explanations and receive responses in the form of clarifications, preferences, corrections and affect. Human–in-the-loop learning paradigms such as preference-based RL and RLHF make this explicit by treating human feedback as the reward signal that shapes a model's policy (Christiano et al., 2017; Wirth et al., 2017; Ouyang et al., 2022; Kaufmann et al., 2023). Similarly, the conversational search and clarification-question literature views users as adaptive feedback sources: systems ask questions to resolve ambiguity and refine intent, and then update retrieval or ranking based on the user's answers (Rao & Daumé III, 2018; Aliannejadi et al., 2019; Krasakis et al., 2020; Salle et al., 2022). More traditional mixed-initiative frameworks also explicitly frame interaction as a joint activity in which human and agent alternate initiative, coordinating to decide who should act next and how (Horvitz, 1999; Ferguson & Allen, 2007).

In all these settings, the user is a structured, partially observable environment whose "state" includes goals, preferences, and mental models, and whose responses provide rich feedback. Treating the user as an environment makes it natural for an agent not only to probe user cognition, but also to infer and update an implicit user profile over time (e.g., stable preferences, expertise level, tolerance for risk, and interaction style) and then *dynamically adjust* its questioning strategy, explanation granularity, and recommendation policy accordingly. Concretely, the agent can ask targeted clarification questions with high expected value of information, adapt policy based on preference signals, or share initiative in collaborative settings. Creativity then emerges at the interaction level: agents can jointly re-frame tasks with users, co-construct novel goals, and discover unconventional solutions (Kumar et al., 2025; Wu et al., 2025), such as through brainstorming or ideation settings where agent participation measurably affects creative outcomes (Wang et al., 2025c; Houde et al., 2025). This is qualitatively different from static prompt–response behavior, and arises precisely because the user is treated as a dynamic, interactive environment.

Case: Other Models as Environment. Modern LLM systems increasingly route queries across multiple models with different capabilities and costs. From the interaction perspective, the pool of available models itself also forms an environment: each "action" is a query to another model, while the "response" is the model's output. Multi-LLM routing frameworks like FrugalGPT (Chen et al., 2024), Router-R1 (Zhang et al., 2025a), xRouter (Qian et al., 2025b), and MasRouter (Yue et al., 2025) explicitly formulate this as a sequential decision problem where a controller learns which models to invoke, in which order, and how to aggregate responses. In evaluation, agent systems may also include auxiliary models that provide feedback directly guiding subsequent calls. For instance, in LLM-as-a-judge setups, a separate evaluator will produce scores, preferences, or critiques that are treated as observations for selecting among candidate generations, making calls to other models an explicit interaction step rather than an offline metric (Zheng et al., 2023; Gu et al., 2024).

In all these cases, the environment becomes a structured ecosystem of models that returns high-level, semantic feedback such as answers, scores, or critiques. Creativity here emerges when the agent does more than statically choose a single expert: it can iteratively query, compare, and arbitrate among different models, discover complementarities (e.g., cheap but strong on specific domains vs. expensive generalists), and adapt routing strategies over time (Chen et al., 2024; Zhang et al., 2025a; Yue et al., 2025). This opens a new axis of creative behavior: inventing novel "model-compositions" and interaction protocols, rather than merely new behaviors within a single fixed model.

Case: Other Agents as Environment. Finally, when multiple agents coexist, each agent's environment includes the others: a message, action, or norm emitted by one agent becomes part of another's observable context. Multi-agent LLM frameworks such as CAMEL (Li et al., 2023) and AutoGen (Wu et al., 2024) explicitly instantiate societies of agents that communicate via natural language to solve tasks cooperatively. Generative Agents (Park et al., 2023) populate an interactive town with LLM-driven characters whose social interactions give rise to emergent coordination and conventions. Multi-agent social-simulation frameworks further treat language-native agents as nodes in a social system to study phenomena such as opinion dynamics and social influence (Gao et al., 2024a; Lin et al., 2025).

These systems show that when agents treat each other as part of the environment, responding to expressed intentions, norms, and actions, complex social behavior may emerge. From the creativity perspective, interaction with other agents enables socially scaffolded novelty: agents can negotiate roles, teach or criticize one another, form committees or debates to explore alternative hypotheses, and converge on solutions that no single agent would find alone (Li et al., 2023; Wu et al., 2024; Park et al., 2023). Empirical studies of LLM-based multi-agent simulations suggest that such systems can reproduce non-trivial social dynamics and generate diverse solution trajectories beyond their training data (Gao et al., 2024a; Lin et al., 2025). Treating "other agents as environment" therefore also exposes a rich space of creative behaviors grounded in social inference, coordination, and emergent collective problem solving.

3.3. Research Direction: Creative Interaction

A direct consequence of "everything is environment" is that **environment choice** becomes a core focus of agency. Future creative agents should therefore be studied as *interaction brokers*: systems that maintain a portfolio of influence channels (user, LLMs, APIs, GUIs, peer agents) and decide which environment to engage next under uncertainty. Recent benchmarks already expose how hard it is to act robustly in realistic interfaces (Xie et al., 2024b; Rawles et al., 2025), but a more radical direction is to let agents discover and recruit new interaction environments midstream, not merely switch among pre-specified ones, thereby expanding the reachable solution space. In this view, creativity is not just novelty within one sandbox, but the ability to recognize that the current environment is limiting, and to actively search for a more leverageable interaction channel (asking the user, consulting a judge, calling another agent expert, or entering a richer interface) at the right time.

A second underexplored frontier is **interface plasticity**: the ability to *reconfigure how interaction happens* by modifying the interface itself (e.g., deciding what information to surface, what constraints to impose, and what protocol to follow). Rather than treating environments as fixed, agents should adapt the interface to fit the task: elicit missing variables, create structured checklists or schemas, and introduce new interaction routines that make downstream reasoning and action easier. This begins to look feasible in recent research: open-action learning targets expanding what an agent can do through experience (Zhao et al., 2024; Qiu et al., 2025; Zhang et al., 2025b), and code-based interaction provides a unifying substrate for iterating over complex multi-step behaviors (Wang et al., 2024b). A natural extension is to make "interaction discovery" explicit: agents should probe unfamiliar settings to infer latent state variables, failure modes, and controllable knobs, then stabilize an effective protocol for subsequent episodes, echoing how persistent interaction patterns emerge in open-ended environments (Wang et al., 2024a).

Finally, **evaluation** must also move beyond "can the agent solve tasks in current environment?" toward "can the agent *discover, choose, and reshape* its environments to solve tasks that would otherwise be out of reach?" Multi-environment benchmarks such as AgentBench (Liu et al., 2024), GAIA (Mialon et al., 2024; Froger et al., 2025) and Tau-Bench (Yao et al., 2024; Barres et al., 2025) already stress that tool use, web interaction, and multi-step decision-making require more than static prompting, while workplace-style suites highlight the importance of mixing browsing, coding, execution, and collaboration in a single coherent setting (Xu et al., 2024). In scientific discovery settings, the same capability should be evaluated not only on solving a given problem, but on problem discovery and method discovery under shifting evidence and constraints. Based on these, the natural next step is to design interaction-switching benchmarks, where agents are rewarded for identifying when the current interaction regime is bottlenecked, justifying an alternative in terms of expected leverage, and demonstrating that the new regime improves what becomes solvable. In short, creative interaction should be measured at the level of what becomes actionable, not only what gets completed.

4. Goal-Oriented Perspective: Everything Is Tool

4.1. Generalizing the Concept

Traditional agent work typically treats tools as *explicit tools*: pre-packaged interfaces (e.g., search engines, calculators, file writers, code executors) with clearly documented inputs/outputs and an intended purpose. At scale, one can also view the web's APIs (weather, booking, payments) as an ever-growing tool library. This paradigm has been productive, as tool use can improve reliability and reduce hallucination by grounding outputs in external functions, but it also quietly fixes the agent's "means space" to what designers have already enumerated and described (Qu et al., 2025; Qin et al., 2024).

From a goal-oriented perspective, this is still too narrow. Human tool use historically did not begin with a catalog of ready-made tools; it began with discovering that ordinary things have attributes that can be exploited for a goal. This motivates a complementary category: *implicit tools*, i.e., entities not designed as tools, whose utility is unlocked by recognizing and leveraging their properties. Thus, we propose a broader definition:

A tool is anything whose attributes can be exploited to advance the agent's goal.

This shifts emphasis from what an object is to what it affords. The intuition is familiar: lacking a bottle opener, a table edge becomes a tool because its rigidity and sharp corner can pry the cap. Classic affordance theory also formalizes this relational view: action possibilities are defined jointly by environmental properties and the agent's capabilities and goals (Gibson, 2014).

This broadened definition also suggests why current agents are not yet "creatively tool-competent." Many systems learn tool use mainly from natural-language tool descriptions or fixed schemas, which encode intended usage but obscure latent, repurposable attributes (Qu et al., 2025; Wang et al., 2025b). Therefore, our position is that the next step is attribute-level utility discovery: agents should learn to search for, hypothesize about, and validate useful attributes in surrounding resources, thereby turning non-tools into tools when needed.

Goal-Oriented View of Creative Agency

A creative agent should treat *all means for goal achievement* as potential tools, and discover their utility at the level of attributes rather than descriptions.

In this way, recognizing 'everything as tool" provides a foundation for creativity from a goal-oriented perspective. Under this view, an agent is creative not because it uses more tools, but because it can discover utility where none was explicitly engineered. Creativity thus arises from the ability to bridge goals and attributes, rather than from adherence to predefined tool boundaries. In the following section, we formalize this intuition by introducing an attribute-matching model of agent cognition, which constitutes a core mechanism underlying creative problem solving.

4.2. How Creativity Emerges: Attribute Matching Model

The goal-oriented view above suggests that creative tool use is not primarily about having a larger tool library, but about *deriving means from attributes*. We model this as an *attribute matching* process: the agent decomposes a goal into attribute-level requirements ("what properties are needed?") and then assigns available entities (objects, APIs, skills) as *means* by matching their attributes to those requirements ("what in my surroundings has those properties?"). Yet current LLMs/VLMs often rely on surface-correlated representations optimized for prediction, rather than explicitly maintaining and testing beliefs about latent functional attributes. Motivated by this gap, we formalize attribute matching under uncertainty and show how it induces goal-conditioned probing.

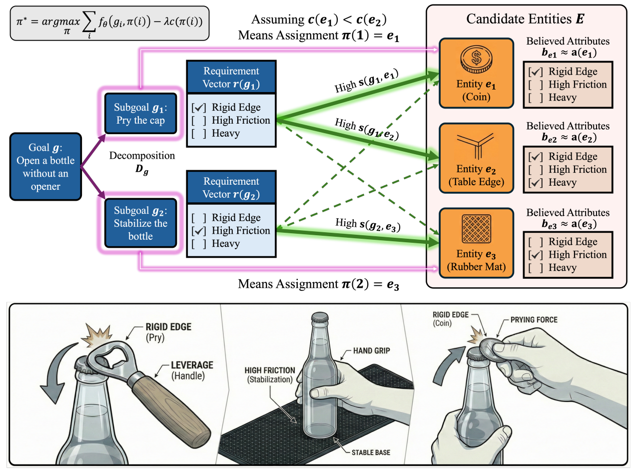

Figure 2. Illustration of the attribute matching model with example.

Entities, attributes, and goal decomposition. Let $\mathcal{E} = \{e_1, \ldots, e_n\}$ be the set of candidate entities the agent can potentially exploit, including explicit tools (e.g., search, code execution), implicit tools (e.g., a table edge), and learned skills (e.g., a reusable script). Let $\mathcal{A}$ be an attribute vocabulary of size K. Each entity e has an (unknown) attribute vector $\mathbf{a}(e) \in \{0,1\}^K$; under partial observability the agent maintains a belief b_e over $\mathbf{a}(e)$. Given a goal g, the agent produces a decomposition $\mathcal{D}(g) = \{g_1, \ldots, g_m\}$ and maps each subgoal g_i to an attribute requirement vector $\mathbf{r}(g_i) \in \{0,1\}^K$.

Matching score and means assignment. For each subgoal–entity pair, the agent computes a compatibility score that estimates how well entity e can serve as the *means* to achieve subgoal g_i. A simple instantiation is

$$s(g_i, e) = \langle \mathbf{r}(g_i), \mathbb{E}_{\mathbf{a}\sim b_e}[\mathbf{a}] \rangle,$$

i.e., the expected overlap between required attributes and the entity's believed attributes; more generally this can be a learned scorer $f_\theta(g_i, e)$. The agent then chooses a *means assignment* $\pi : \{1, \ldots, m\} \to \mathcal{E}$, where $\pi(i)$ denotes the entity selected to serve subgoal g_i, by trading off match quality and cost:

$$\pi^\star = \arg\max_{\pi} \sum_{i=1}^{m} \Big(f_\theta(g_i, \pi(i)) - \lambda\, c(\pi(i)) \Big).$$

Example Case. As shown in Figure 2, for "open a bottle without an opener," the agent may decompose into g_1 = "pry the cap" and g_2 = "stabilize the bottle." It may set $\mathbf{r}(g_1)$ to require *rigid edge*, and $\mathbf{r}(g_2)$ to require *high friction*. Then a coin or table edge becomes a strong candidate for $\pi(1)$, while a rubber mat becomes a strong candidate for $\pi(2)$. The creative step is that these entities are selected *because of their attributes*, not because they were labeled as bottle-opening tools.

Interaction as attribute inference. In practice, attributes are uncertain (the edge may be too rounded; an API may not support a parameter), so matching induces goal-conditioned probing. Let q denote an interaction (a test call, a small probe action, or a clarification question) that updates beliefs b_e. The agent can select q to maximize the expected improvement in achievable matching:

$$q^\star = \arg\max_q \ \mathbb{E}\Big[\max_\pi \sum_i f_\theta(g_i, \pi(i)) \ \Big| \ b, q\Big] - \eta \, \mathrm{cost}(q).$$

This captures why creativity is inherently interactive: the agent does not merely *assume* utility from a description, but *tests* hypothesized utility at the attribute level.

4.3. Research Direction: Creative Tool Use

Treating "everything as tool" implies that robust agency depends on **attribute-level substitution**: when the intended object is missing or fragile, the agent should re-derive means from whatever attributes the environment can supply. This shifts tool use from picking items in a catalog to running a goal-conditioned loop that hypothesizes candidates, probes the attributes that matter, and replans when evidence contradicts assumptions. For instance, if an "extract structured data" API fails, a creative agent should recognize that the goal only requires attributes like repeatability and structured output, pivot to a GUI pathway, and validate it via small schema checks. The underlying insight is affordance-theoretic: utility arises from the fit between goal requirements and environmental properties rather than from designer labels (Gibson, 2014). Methodologically, this motivates training and evaluation that reward "hypothesize $\rightarrow$ test $\rightarrow$ adapt" behavior under partial observability, beyond fixed-schema tool invocation (Qu et al., 2025; Qin et al., 2024; Wang et al., 2025b).

A second direction is to make creative tool use cumulative by operationalizing **inventability**: once an attribute match is validated, the agent should compile the procedure into a reusable operator with explicit inputs/outputs, preconditions, and failure detectors. This is not just engineering convenience but a cognitive claim: current big-data training paradigms often overfit to word-/pixel-level correlations, transfer poorly across functional similarity or affordances, and seldom turn interaction experience into compositional, long-horizon routines. Inventability addresses this by converting discovered *attribute-level strategies* into explicit operators, so future tasks start from a richer action space rather than repeating exploratory search. For example, an agent that repeatedly evaluates claims by combining browsing, extraction, and code-based consistency checks should synthesize a stable evaluator action chain that can be called, audited, and revised. This aligns with open-action learning that targets creating new callable actions from experience (Zhao et al., 2024; Qiu et al., 2025) and code-centric action spaces that naturally support composition and refactoring (Wang et al., 2024b). All of these imply that reliability should be grounded in verification and provenance of invented operators instead of semantics of pre-described tools.

Finally, beyond substitution and inventability lies **tool shaping**: creative agents should be able to transform available resources so their attributes become suitable for the goal. In many realistic failures, the environment contains near-matches that miss a decisive property such as stability, structure, or controllability, and creativity consists in adding that missing attribute rather than abandoning the candidate. For example, when an API is unreliable, an agent can introduce validation, retries, and fallbacks to turn it into a dependable operator; when model outputs are expressive but inconsistent, the agent can impose a typed interface and lightweight checks so free-form generation becomes callable. This treats affordances as engineerable, not merely discoverable, and motivates learning *attribute dynamics*: which interventions predictably improve which properties, at what cost, and with what side effects. Evaluation should therefore include settings where the strongest solution requires reshaping the tool itself, rewarding durable transformations that transfer across tasks rather than one-off workarounds.

5. Discussions

Previous research directions and our positions further inspires some interesting topics, as listed in the following.

Self-Affordance. If other agents can be viewed as the environment, an agent should also model *itself* as a structured environment, especially its own affordances. In multi-agent settings, this suggests learning a *comparative-advantage* profile: what the agent reliably does better than peers, when those strengths apply, and where it should defer. Today such "self-affordance" is largely hand-specified (e.g., via system prompts), but a more scalable direction is to acquire and update it through experience under collaboration and competition, consolidating it into a persistent, queryable profile. Key open questions include how to represent self-affordance (e.g., as uncertainty-aware capability manifolds), how to quantify it from outcomes and feedback, and how to make it legible and usable for coordination (so that others can route sub-tasks to the right agent).

Tool Composition. Tool inventability (as mentioned in Section 4.3) alone may be insufficient because tool spaces are inherently hierarchical: atomic actions compose into reusable macros, and macros further compose into higher-level programs. This hierarchy introduces new failure modes and opportunities: composing tools can yield emergent affordances (not merely additive), but also compounding errors and brittle dependencies. A central research challenge is reliable composition, including verification of feasibility, re-estimation of composed-tool affordances under context shift, and detection of unintended side effects. Composition also stresses decision policies: as dimensionality grows, attribute matching process must also become more efficient, deciding when to invoke first-order tools versus higher-order composites, and when to refine or adapt composites online.

Domain Transfer. Although our case studies in Section 3 emphasize digital environments (e.g., software APIs or other models) and our goal realization examples in Section 4 draw from the physical world, the underlying principle is domain-general: agents must (i) actively model heterogeneous environments they can interact with, and (ii) match task requirements to environment-specific affordances. In physical settings, this means modeling people, institutions, and spaces as distinct environments with profiles, choosing whom/what to engage, and integrating feedback along the interaction trajectory. In software settings, the same logic applies: collaborating agents and tools carry specialized capabilities (e.g., math vs. coding), so robust performance requires dynamically decomposing tasks and matching subproblems to the best affordances. Understanding what transfers, what must be re-identified, and how to unify representations across domains remains an important direction.

6. Call For Actions

Creativity in agents is best viewed as a trajectory-level property: not randomness in a single completion, but the ability to keep several viable trajectories alive long enough to compare, test, revise, and recombine them (Vijayakumar et al., 2016; Holtzman et al., 2020). This "structured divergence" is precisely what long-horizon tool use demands: agents succeed by proposing alternatives, backtracking when a path fails, and exploiting rare-but-valid affordances (Yao et al., 2023; Wang et al., 2024a).

This immediately implies that creativity is a pipeline problem with two coupled steps: **proposal** (can the model generate diverse, coherent candidates?) and **selection** (does the model preserve that diversity, or collapse to a default mode?). Pre-training and post-training exert pressure on different steps of this same pipeline. Pre-training largely shapes what the model can readily *propose*; post-training largely shapes what the model is willing to *commit to*. Failures in either stage may reduce creative capacity.

Two linked failure modes. *Pre-training can narrow proposals.* Next-token prediction rewards faithful regeneration of observed text and can behave like trajectory cloning, favoring typical continuations over alternative decompositions, counterfactuals, and unconventional tool uses. *Post-training can over-collapse selection.* RL/RLHF and related preference objectives improve instruction-following (Ouyang et al., 2022; Rafailov et al., 2023) but often increase mode-seeking and reduce behavioral diversity (Kirk et al., 2024; Yun et al., 2025); KL-style constraints can further sharpen the output distribution toward dominant modes (Wang et al., 2025a). For agents, this combination is particularly damaging: a narrow proposal set plus aggressive selection eliminates the exploration needed for creativity.

Action 1: Diagnose proposal vs. selection. We need evaluations that tell *where* creativity fails. Sometimes the model never produces genuinely different options (a proposal failure). Other times it can produce them, but it keeps collapsing to the same safe/default choice (a selection failure). A simple probe is **branch-before-commit**: force the model to write several different plans first, then choose one. We then score (i) whether the plans are meaningfully different in assumptions, subgoals or tool use, and (ii) whether the final choice consistently ignores high-quality but less "default" options. In this framing, scaffolds like explicit branching become tests that reveal whether we should fix the generator (pre-training/proposal) or the chooser (post-training/selection).

Action 2: Train calibrated exploration across phases. On the *pre-training* side, the goal is to broaden the **proposal distribution**: explicitly train the model to generate multiple distinct but valid continuations/decompositions for the same context (e.g., multi-continuation targets, counterfactual rewrites, alternative plans), so diversity is a learned capability rather than a sampling artifact. On the *post-training* side, the goal is to preserve that diversity at **selection time**: reward correct low-probability trajectories, downweight repeatedly chosen dominant solutions, and tune KL strength based on observed concentration. This is consistent with evidence that explicitly rewarding less-likely but correct solutions mitigates mode dominance (He et al., 2025).

Action 3: Evaluate capability frontiers. Finally, creativity and "upper-bound capability" should not be inferred from expensive sampling alone. We call for frontier-style **metrics** that, under a fixed compute budget, measure (i) how many distinct solution modes the model can reach (coverage) and (ii) how quickly it collapses onto a few favorites (concentration). Within limited search budgets, these metrics can detect when higher preference come at the cost of losing long-tail behaviors that support creative agency.

7. Alternative Views

A common critique to our position might be that creativity is merely an artifact of scale, like larger pretrained models, or of accumulating specialized skills, rather than any universal interpretive principle. We argue instead that skill libraries without open-ended reinterpretation resemble industrial automation: highly capable, yet fundamentally non-creative. Creativity does not emerge from *more* tools alone; it emerges from a policy that can *reinterpret* what counts as a tool, i.e., from seeing actionable affordances broadly and revising that view as context changes.

Another concern is definitional: if everything can be viewed as both environment and tool, what distinction remains? Our position is that the difference is perspectival rather than ontological. From an interaction perspective, we call something an *environment* when it receives actions and returns feedback; creativity then hinges on choosing which environments to engage and how to adapt based on their responses. From a goal-oriented perspective, we call something a *tool* when it is treated as a means with specific affordances; creativity then hinges on decomposing goals into attributes and re-matching them to available affordances. Many entities, such as users or other agents, naturally admit both views, and which label is appropriate depends on whether we emphasize adaptive interaction or goal realization.

8. Conclusion

In this paper, we highlight that agency should not be reduced to predefined APIs or fixed action spaces: an agent is defined by goal-directed interaction, and its creative power grows when it can opportunistically reinterpret what surrounds it. Concretely, we advance two linked positions: **everything is environment** (agents should treat any influenceable surrounding as a candidate interaction space) and **everything is tool** (agents should treat any exploitable resource as a potential means, discovered at the level of attributes rather than labels). We call for methods and evaluations that reward this kind of open-ended environment choice, affordance discovery, and calibrated exploration instead of premature behavioral collapse.

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